NeuroImage 300 (2024) 120850

Contents lists available at ScienceDirect

NeuroImage

journal homepage: www.elsevier.com/locate/ynimg

Ultra-high-resolution mapping of myelin and g-ratio in a panel of *Mbp* enhancer-edited mouse strains using microstructural MRI

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ARTICLE INFO

Keywords:
Myelin water imaging
Gradient echo MRI
Quantitative MRI
Myelin basic protein
Mouse models

ABSTRACT

Non-invasive myelin water fraction (MWF) and g-ratio mapping using microstructural MRI have the potential to offer critical insights into brain microstructure and our understanding of neuroplasticity and neuroinflammation. By leveraging a unique panel of variably hypomyelinating mouse strains, we validated a high-resolution, model-free image reconstruction method for whole-brain MWF mapping. Further, by employing a bipolar gradient echo MRI sequence, we achieved high spatial resolution and robust mapping of MWF and g-ratio across the whole mouse brain. Our regional white matter-tract specific analyses demonstrated a graded decrease in MWF in white matter tracts which correlated strongly with myelin basic protein gene (*Mbp*) mRNA levels. Using these measures, we derived the first sensitive calibrations between MWF and *Mbp* mRNA in the mouse. Minimal changes in axonal density supported our hypothesis that observed MWF alterations stem from hypomyelination. Overall, our work strongly emphasizes the potential of non-invasive, MRI-derived MWF and g-ratio modeling for both pre-clinical model validation and ultimately translation to humans.

1. Introduction

Myelin is an insulating material formed from the plasma membrane of oligodendrocytes in the CNS and Schwann cells in the PNS. It is spirally wrapped and tightly compacted around axons where it plays a fundamental role in enhancing neuronal conduction velocity. It is now increasingly recognized that myelin plays a prevalent role in axonal metabolic support (Moore et al., 2020) and that it undergoes plastic changes in the mature CNS with significant functional consequences (Xin and Chan, 2020). Given the important functions assigned to myelin, non-invasive imaging approaches coupled with catered biophysical models of CNS microstructure have the potential to advance our understanding of neurodevelopment, neuroplasticity, and demyelinating neurological conditions.

Currently however, magnetic resonance imaging (MRI)-derived, non-invasive estimates of myelin volume fraction (MVF), axon volume fraction (AVF) and g-ratio (axon diameter divided by total fiber diameter) rely on assumptions related to axon geometry and circular layering of the myelin phospholipid bilayer around axons (Jung et al., 2018; Stikov et al., 2015). While these models have been applied in animal models and humans, they suffer from several important constraints, the most notable being lack of robust calibration. No study has robustly validated MVF, AVF and g-ratio mapping in graded hypomyelinating models against MRI-derived myelin signal. Second, the only MVF validation study on a gene knockout model with varying levels of myelination to date included measurements spanning only the corpus callosum and the temporal limb of the anterior commissure (West et al., 2018b). Our study examines tracts that span the whole brain along the

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https://doi.org/10.1016/j.neuroimage.2024.120850
Received 2 April 2024; Received in revised form 27 August 2024; Accepted 9 September 2024
Available online 10 September 2024

rostro-caudal axis. If the myelin volume fraction mapping framework is to become more widely utilized for human imaging, such a whole-brain analysis is critical. Thus, our study evaluates the impact of differential *Mbp* expression on myelin elaboration in multiple myelinated tracts in the mouse brain. Third, the only existing high resolution mouse brain imaging study of hypo- and hypermyelinating mouse lines used a gadolinium contrast agent to enhance the MRI signal derived from myelin (West et al., 2018a). While this practice is common for preclinical studies of volumetry in knockout models, it alters MRI relaxation rate constants (Barrett et al., 2023; Iyad et al., 2023). Thus, it artificially changes MRI-derived MVF, AVF and g-ratio estimates in CNS tissue. In this work, we performed measurements of MVF, AVF and g-ratio mapping without employing contrast agents. Consequently, all biophysical measurements of myelin, axons, and MRI-derived g-ratio were undertaken using the native, *endogenous* MRI signal. Finally, despite recent advances in accelerated MWF (Dvorak et al., 2023; Gong et al., 2023; Lee et al., 2024) and aggregate g-ratio (Cortina et al., 2022) acquisition schemes, there does not exist a consensus regarding which method, or combination of methods, is most amenable for fast and efficient clinical implementation. The acquisition method and processing pipeline we propose in this work addresses this issue.

For our imaging experiments, we exploited a panel of mice that elaborate and maintain graded levels of CNS myelin due to edits introduced into transcription enhancers that drive expression of the *Mbp* gene (Bagheri et al., 2020). Under normal physiological conditions, myelin is densely compacted around the axon with the aid of myelin-associated proteins and has a water content of ~40 % in-situ, with dry mass proportions of ~70 % phospholipid to ~30 % protein. Proteolipid protein (PLP) and myelin basic protein (MBP) are two main contributors to the myelin protein mass and participate in myelin elaboration and stabilization (Morell, 1999). However, the precise impact of removing MBP on phospholipid bilayer stabilization and myelin integrity across the whole brain has not been evaluated using brain imaging. In this regard, the present work uniquely achieved the first whole mouse brain white matter characterization of graded hypomyelination, resulted by the deletion of one or more of the *M1E, M3*, and *M5* transcription enhancers of *Mbp* (Bagheri et al., 2020). We leveraged these well-characterized hypomyelinating mouse models as a testing ground to investigate the sensitivity and specificity of our myelin volume fraction, axon volume fraction and g-ratio mapping paradigm. Throughout this manuscript, MBP refers to the myelin basic protein, whereas *Mbp* denotes the gene encoding this protein.

The first critical step in our high-resolution imaging assay was to estimate myelin volume fraction. For this purpose, we employed myelin water imaging (MWI). MWI is a magnetic resonance imaging (MRI) technique that is sensitive to water protons residing between the phospholipid bilayers of the myelin sheath. MWI can be implemented by exploiting the differential T_1 (Labadie et al., 2014) and T_2 (Nguyen et al., 2016; Whittall and MacKay, 1989) relaxation properties of myelin water protons, or a combination thereof (Bouhrara and Spencer, 2017; Deoni et al., 2008). In the context of transverse relaxometry, myelin water protons are particularly distinguishable from intracellular/extracellular water (IEW) protons by their characteristically short T_2 (MacKay et al., 1994). T_2-based MWI measurements in the brain have traditionally been conducted using multi-echo spin echo (MESE)-based measurements of the T_2 decay curve sampled at each MRI voxel (Alonso-Ortiz et al., 2015). The decay data can then be decomposed into a spectrum of logarithmically spaced, non-orthogonal exponential functions, with contributions from myelin water and IEW protons, using a regularized non-negative least squares (NNLS) algorithm (MacKay and Laule, 2016). The ratio of the T_2 spectrum corresponding to myelin water, obtained via short T_2 thresholding, to the total spectral density is termed the myelin water fraction (MWF). NNLS-derived MWF has been shown to correlate well with myelin histology measures. However, the NNLS fitting procedure is computationally expensive and ill-conditioned, resulting in MWF maps that can be noisy and difficult to interpret. Additional technical challenges related to MESE acquisitions can include long scan times, potential for prohibitively high specific absorption rate (SAR) at high field due to the use of refocusing pulses, and necessary corrections for stimulated echoes and transmit field (B_1^+) inhomogeneity (Lee et al., 2021).

Similar multi-exponential relaxation behavior of myelin water can be efficiently measured using the apparent transverse relaxation constant, T_2*. T_2* relaxation times in human tissue are shorter due their sensitivity to magnetic field inhomogeneities (Chavhan et al., 2009). Using a multi-echo gradient-recalled echo (mGRE) sequence and a simplified three-pool exponential model of white matter incorporating myelin, IEW, and mixed water pools, Du et al. (Du et al., 2007) produced the first MWF maps exploiting differential T_2* relaxation properties of myelin. While this method produced biologically plausible MWF images and was sensitive to multiple sclerosis (MS)-linked demyelination in fixed tissue, it required careful parameter selection to seed the initial conditions for the non-linear least squares (NLLS) algorithm. This is challenging because it is difficult to ascertain NLLS initial conditions a priori (Liu et al., 2022). mGRE-based MWF acquisitions have several advantages, including increased scan efficiency and lower SAR. The increased scan efficiency can be leveraged for conducting ultra-high resolution microstructural imaging to measure MWF. However, mGRE readouts of MWF can also come at expense of lower signal-to-noise ratio (SNR) and higher sensitivity to static magnetic field inhomogeneities (ΔB_0 effects, Alonso-Ortiz et al., 2018). Bipolar readout gradients applied during mGRE acquisitions may additionally lead to misalignment of odd and even echoes. Thus, there remains a critical need for accurate mGRE artifact correction strategies, as well as model-free, computationally expedient algorithms for producing accurate mGRE MWF maps. To that end, a blind source separation algorithm for mGRE-based MWF mapping was initially developed by Song et al. (Song et al., 2020). This algorithm, which we will refer to as linear dimensionality reduction with robust principal component analysis (LDR-rPCA), computes MWF maps by separating the slowly decaying and rapidly decaying T_2* components corresponding to the IEW and myelin water proton pools. LDR-rPCA satisfies many of the requirements of a robust T_2*-based MWF mapping technique. It is fast and does not necessitate a priori numerical or biophysical models of T_2* relaxation. Thus, the output MWF maps are obtained in a wholly data-driven manner. As part of this work, we (i) implemented a catered bipolar readout gradient correction designed for ultra-high resolution microstructural imaging and (ii) analyzed our mGRE decay data using an implementation of LDR-rPCA engineered for sensitively mapping the short T_2* (<25 milliseconds) myelin water signal.

To extend beyond MVF alone and model g-ratio in the brain, MRI-derived myelin mapping approaches are typically combined with an MRI contrast sensitive to axonal volume fraction (Campbell et al., 2018; Mohammadi and Callaghan, 2021). The applications of such MRI g-ratio imaging include monitoring of neurodevelopment in children (Dean et al., 2016), healthy aging in adults (Bouhrara et al., 2021), and disease progression in relapsing-remitting MS (RRMS) (York et al., 2021). The axon area-weighted mean g-ratio within an imaging voxel can be estimated by separately sensitizing the MR signal to myelin water and axon water compartments (West et al., 2016). Assuming a myelin-insensitive acquisition, the axon water fraction (AWF) can be obtained from diffusion-weighted MRI data via biophysical models such as neurite orientation dispersion and density imaging (NODDI, Zhang et al., 2012). In g-ratio mapping approaches, the MWF value, which provides only indirect information about myelin volume within a voxel, is typically scaled to myelin volume fraction (MVF) based on assumptions about myelin mass density (West et al., 2018b), or the geometric properties of myelin (Jung et al., 2018). Following co-registration of MVF and AWF maps, these are integrated via an analytical framework to yield the aggregate g-ratio of all axons within a voxel (Stikov et al., 2015). Our present study combined MRI-derived myelin volume fraction mapping with axon water fraction mapping. This allowed high resolution MRI g-ratio estimation in our panel of variably hypomyelinating mice.

To summarize, in this work, we combined a highly novel ultra-high resolution MRI acquisition and MWF calculation scheme to characterize the impact of *Mbp* expression on whole mouse brain hypomyelination. Importantly, our hypomyelinated models served as a sensitive testing ground for calibrating MRI-derived myelin volume fraction and g-ratio relative to *Mbp* mRNA accumulation and immunohistochemistry (IHC)-derived MBP content. We additionally augmented these findings with diffusion MRI-derived AWF and light chain neurofilament (LNF) IHC measurements, thereby demonstrating that variations in LDR-rPCA MWF are driven predominantly by variable myelin elaboration. Our MRI paradigm has the potential to provide a highly innovative approach for non-invasive microstructure mapping for both whole-brain genotype-phenotype relations in mice and in the human CNS.

2. Theory

2.1. Bipolar gradient odd/even echo misalignment

During the acquisition of 3D mGRE images, *k*-space may be populated in an accelerated manner with the aid of bipolar readout gradients. Bipolar readouts offer increased acquisition efficiency compared to monopolar acquisitions (Eckstein, 2019). However, phase differences accumulated in *k*-space due to gradient delays and eddy currents can contribute to misalignment in the magnitude image between odd and even echoes in image-space (Cohen-Adad, 2014). This misalignment of the underlying anatomy in the echo train can further propagate as errors in downstream model fitting. Echo misalignment errors are accentuated at interfaces between different tissue types, where partial volume effects may exist. To address the echo misalignment in this work, we proposed an empirical *k*-space based correction. The method is inspired by the phase image misalignment correction strategy previously proposed by Lu et al. (Lu et al., 2008). However, our approach was modified to correct multi-echo magnitude images.

Our correction strategy is defined as follows: Let $K_O(k_x, k_y, k_z)$ and $K_E(k_x, k_y, k_z)$ represent the first odd and even complex *k*-space volumes within the mGRE train, respectively. Assuming the phase encoding direction is along the *x*-axis, the phase shift of even-numbered echoes relative to odd echoes can be modelled as:

$$K_O(k_x,\ k_y,\ k_z) \approx\ K_E(k_x,\ k_y,\ k_z)\ \cdot\exp(-jk_x\Delta_G) \tag{1}$$

where Δ_G represents the phase shift in *k*-space and $T_2{}^*$ relaxation is ignored. When the magnitude image echo train is reconstructed with a 3D inverse Fourier transform, the even echo magnitude image, I_E, is shifted along the *x*-axis:

$$|\boldsymbol{F}^{-1}(K_E(k_x,\ k_y,\ k_z)\cdot\exp(-jk_x\Delta_G)\ )| = I_E(x+\Delta_G,\ y,\ z) \tag{2}$$

However, the contribution of Δ_G in image space may displace even echoes by an amount smaller than the prescribed resolution, resulting in sub-voxel shifts. This leads to a characteristic "jitter" of the mGRE magnitude decay curve along the T_E-axis. To overcome this problem, we applied an opposing phase ramp $exp(+jk_x\beta)$ to even echoes in *k*-space. Since the quantity Δ_G is not known a priori and may vary between samples due to the interaction of the sample with the static magnetic field, the slope of the opposing ramp β was determined empirically. This was carried by iterating over several β values and computing the mutual information (MI) between odd and even echo magnitudes in image space. The value β^* that maximizes the image space MI was then selected and we applied it as a phase ramp to all even echoes in k-space:

$$\mathbf{argmax}_x[\mathbf{MI}(I_O(x),\ I_E(x+\Delta_G-\beta))] := \ \beta^*\ \text{s.t.}\{\beta^*\approx\Delta_G\} \tag{3}$$

An advantage of the method described above is that it does not rely on explicit ΔB_0 modelling and instead corrects for inter-echo misalignment in a data-driven manner. This step allowed more robust measurement of MWF free from anatomical misalignment in consecutive magnitude images in the echo train.

2.2. Implementation of LDR-rPCA and computation of MWF

Below we describe our implementation of LDR-rPCA for ultra-high resolution preclinical imaging. Following echo misalignment correction, the MWF map is computed from the corrected mGRE data, M, using the linear dimensionality reduction with robust principal component analysis (LDR-rPCA) method first introduced by Song et al. (2020). LDR-rPCA computes MWF maps by first separating the slowly decaying and rapidly decaying $T_2{}^*$ components of the mGRE signal originating from the intracellular/extracellular water (IEW) proton and myelin water proton pools. These are termed L_1 and L_2, respectively. Separation of the components is accomplished through singular value decomposition (SVD). A mono-exponentially decaying profile, or "unit-rankness" in L_1 and L_2 is enforced through Hankelization of the mGRE signal prior to decomposition. The final MWF map is computed by taking the ratio of L_2 and the sum of L_1 and L_2 at the first echo time TE_1. Advantageously, artefactual components that do not exhibit a mono-exponential decay profile following Hankelization are isolated in the sparse component S. This provides an inherent denoising of the resultant MWF maps.

The LDR-rPCA algorithm can be broadly divided into two steps: (i) an initialization step and (ii) a minimization step. In the initialization step, a low rank approximation of the Hankelized source mGRE signal, M, is obtained through non-negative matrix factorization (NMF) under the assumption that inter-echo spacing, ΔTE, is constant for all echoes:

$$W,\ H = NMF(\mathscr{H}_l(M)) \tag{4a}$$

Here, $\mathscr{H}_l$ is the Hankelization operator with length l, W is the temporal basis matrix, and H is the spatial weight matrix. Unlike canonical PCA, NMF ensures that the components of the basis matrix are strictly non-negative and, therefore, plausible in the context of MRI. The initial estimate of the slowly decaying low-rank component, L_1, is obtained from the product of the first (largest) temporal basis component and first spatial weight vector (Eq. (4b)). The quickly decaying low rank component L_2 is initialized by subtraction of L_1 from M (Eq. (4c)):

$$L_1 = \mathscr{H}_l^{-1}(W[:,\ 1]\times H[1,:]) \tag{4b}$$

$$L_2 = M - L_1 \tag{4c}$$

The sparse component S is initialized to zero. The initial estimates of L_1, L_2, and S are then updated iteratively by minimizing the convex objective function:

$$J(L_1,\ L_2,\ S) = \ \frac{1}{2}\|\ L_1 +\ L_2 + S - M\ \|_2^2$$
$$+\ \mu_1\sum_r||\mathscr{R}_r(\mathscr{H}_l(L_1))||_* \ +\mu_2\sum_r||\mathscr{R}_r(\mathscr{H}_l(L_2))||_*\ +\ \rho||S||_1 \tag{5}$$

In Eq. (5) $\mathscr{R}_r$ denotes the extraction of locally low-rank patches within each Hankelized matrix at the r^{th} patch. μ_1, μ_2, and ρ are regularization constants, and $||\bullet||_*$, $||\bullet||_1$ are the nuclear norm and l_1 norm, respectively. Eq. (5) is minimized iteratively using the alternating direction method of multipliers (ADMM), whereby the associated augmented Lagrangian function of $J\{L_1, L_2, S\}$ is split into sub-problems $\{U_1, U_2, U_3\}$ with closed-form solutions. The update rules for U_1 and U_2 incorporate singular value decomposition and thresholding of Hankelized patches of L_1 and L_2, thereby enforcing local low-rankness.

Sparsification of S is enforced by implementing soft thresholding in the update rule for U_3. In this way, the estimates for $\{L_1, L_2, S\}$, the associated Lagrangian multipliers $\{Z_1, Z_2, Z_3\}$ and the corresponding sub-problems $\{U_1, U_2, U_3\}$ are updated at every iteration k of the algorithm. The convergence of ADMM is formulated to occur when the relative residual l_2-norm of $\{L_1, L_2, S\}_{k+1}$ and $\{L_1, L_2, S\}_k$ is smaller than a threshold value ε, indicating that L_1 and L_2 are locally low-rank within all patches and additional iterations do not improve their estimate. Finally, the MWF is computed as the ratio of the quickly decaying component L_2 to the sum of L_1 and L_2 at the first echo time (TE_1):

$$MWF(x, y, z) = \frac{L_2(x, y, z, TE_1)}{L_1(x, y, z, TE_1) + L_2(x, y, z, TE_1)} \quad (6)$$

The complete initial implementation of the LDR-rPCA algorithm and its derivation are provided in the original article by Song et al. (2020).

3. Materials and methods

3.1. Tissue preparation

All experiments were carried out in accordance with the guidelines of the Canadian Council on Animal Care. Animal ethics protocol number #2016–7832 was approved by the McGill University DOW Facilities Animal Care Committee. For brain tissue preparation, all mice were anaesthetized with Avertin and transcardially perfused with 10 ml of Hank's balanced salt solution, followed by 10 ml of fixative (2.5 % glutaraldehyde and 4 % paraformaldehyde) at 1 ml/min. Brains were dissected from the skull and were kept in fixative overnight at 4 °C, followed by 7 daily washes in phosphate buffered saline before undergoing magnetic resonance imaging. Twenty-three brains from seven mouse lines which exhibited a broad range of myelination levels were investigated at postnatal day 30 (P30). For the single (*1xKO*)-, double (*2xKO*)- and triple(*3xKO*)-knockout (KO) lines, CRISPR/Cas9 was employed to delete one or more of the *M1E, M3*, and *M5* transcription enhancer sites of the *Mbp* gene (Bagheri et al., 2020). As the *M3KO* and *M5KOΔ3*.6 kb lines exhibited almost identical *Mbp* mRNA transcription profiles, they were treated as an identical group for the purpose of this study. The *Shiverer* line, which is null for *Mbp* and lacks compact myelin, was also included to establish the lowest detectable myelin water signal (Readhead and Hood, 1990). Finally, a knock-in (*KI*) line exhibiting considerably increased *Mbp* mRNA transcription was also included to investigate putative hypermyelination (Bagheri et al., 2024). The number of brains used from each mouse line and the corresponding *Mbp* mRNA measurements derived from quantitative reverse transcriptase polymerase chain reaction (qRT-PCR) of the whole spinal cord are detailed in Table 1.

3.2. MRI acquisition

All imaging was performed using a 7 Tesla Bruker (Billerica, MA, USA) Pharmascan preclinical MRI scanner. For signal excitation and reception, we employed a single channel, cylindrical transceiver coil. For imaging, mouse brains were placed in a receptacle filled with Sigma-Aldrich (St. Louis, MO) Fluorinert FC-40 to provide a signal-free background. T_2*-weighted images were then acquired using a 3D mGRE sequence with TE/ΔTE/TR = 2/2/100 ms, 24 echoes, flip angle = 22°, bipolar readout, 18 signal averages, receiver bandwidth = 100 kHz, matrix size = 135 × 106 × 120 and acquired spatial resolution = 100 × 100 × 100 μm^3. The total scan time for the mGRE acquisition was ~6 hrs. Diffusion-weighted images were acquired using a custom diffusion-weighted gradient and spin echo (dwGRASE) sequence (Wu et al., 2013) with TE/TR = 35/1000 ms, 2 signal averages, RARE factor = 4, gradient pulse duration (δ) = 8 ms, diffusion time (Δ) = 12 ms, prescribed *b*-values = 2500s/mm^2 (30 directions) and 4000 s/mm^2 (60 directions), six $b = 0$ images, matrix size = 90 × 64 × 80 and acquired spatial resolution = 150 × 150 × 150 μm^3. The total scan time for the collection of the diffusion-weighted images was ~13 h. Radiofrequency excitation homogeneity was verified by implementing the double angle method (Stollberger and Wach, 1996), **Figure S1**.

Table 1
The relative *Mbp* mRNA ± the standard error of the mean (SEM) at P30 and number of brains from each mouse line used in this study. A short form for each mouse line was devised to simplify the formal genetic nomenclature in the context of this manuscript.

Mouse Strain	*Mbp* mRNA (% ± SEM)	No. of Brains	Short Form
M3KO;389bpKI	160 ± 6	3	*KI*
Wild Type	100 ± 6	5	*WT*
M3KO / M5KOΔ3.6kb	63 ± 4 / 63 ± 2	3 / 3	*1xKO*
M3M5KO	28 ± 1	3	*2xKO*
M1EM3M5KO	15 ± 1	3	*3xKO*
Shiverer	0 ± 0	3	*Shi*

3.3. MRI data analysis

All data analysis was performed offline using Python 3.8. Complex mGRE images were Fourier reconstructed at native resolution and corrected for sub-voxel misregistration between odd and even echoes along the readout dimension, according to Eq. (3). The phase ramp slope parameter β was varied between ±0.03 in steps of 0.005 to ensure an accurate sampling of possible corrections. Following echo re-alignment, mGRE magnitude images were corrected for Gibbs ringing using the sub-voxel shift method proposed by Kellner et al. (Kellner et al., 2016), implemented in Dipy (Garyfallidis et al., 2014).

MWF maps were then computed using an in-house implementation of LDR-rPCA. The initialization step (Eq. (4a)) was carried out using an NMF algorithm implemented in the NIMFA Python package (Zitnik, 2012). NMF was specifically implemented with 3 temporal bases representing the quickly decaying, slowly decaying, and artefactual components. The NMF was initialized with double singular value decomposition (NNDSVD) seeding and proceeded according to a Euclidean update rule. As per Song et al., the Hankelization length was set to half the number of echoes in the mGRE train, i.e., 12. During the minimization step (Eq. (5)), the regularization constants μ_1, μ_2, and ρ were all set to 1. The regularization constants δ_1, δ_2, and δ_3 for sub-problems $\{U_1, U_2, U_3\}$ were set to 25, 25, and 50 respectively, and the window size of the patch extraction operator $\mathscr{R}$ was set to a random integer between 2 and 8 in each image-space dimension (*x, y, z*). This was done to ensure expedient convergence, as well as adequate coverage when the number of voxels along a given dimension was odd valued. The threshold value ε was set to 0.1, indicating that the l_2-norm of $\{L_1, L_2, S\}_{k+1}$ and $\{L_1, L_2, S\}_k$ changed by a 10th of a percent between iterations at convergence. The MWF was computed as per Eq. (6).

Raw diffusion-weighted GRASE images were corrected for Gibbs ringing and denoised using the Marcenko-Pastur PCA algorithm (Veraart et al., 2016), implemented in Dipy. The NODDI model was implemented using the AMICO software (Daducci et al., 2015). A dot compartment was included to model isotropic restricted diffusion (Tax et al., 2020). The values of intrinsic diffusivity (D_{in}) and isotropic diffusivity (D_{iso}) in the NODDI model were set to 0.9 × 10^{-3} mm^2/s and 2.0 × 10^{-3} mm^2/s, respectively. This was in accordance with the NODDI model implementation of Wang et al. for ex vivo mouse brain imaging (Wang et al., 2019). The axon water fraction (AWF) was computed according to the NODDI formalism as $AWF = v_{ic} \times (1 - v_{iso}) \times (1 - v_{dot})$, where v_{ic}, v_{iso}, and v_{dot} are the volumes intracellular, isotropic, and dot compartments, respectively.

A rigid transform mapping the average of the $b = 0$ dwGRASE images to the average of the mGRE images was used to align the AWF maps to the MWF maps for every mouse brain data set acquired using 7T MRI. To compute the g-ratio, MWF maps were first scaled to myelin volume fraction (MVF) maps using the mass density approach described by West et al. (2018a,b). Values implemented for the myelin and non-myelin water proton pools were those obtained by Jung et al. (2018) from regression analysis of MRI- and electron microscopy-derived measures of myelin volume:

$$MVF = \frac{MWF \times v_{W,NM}}{MWF \times (v_{W,NM} - v_{W,M}) + v_{W,M}} \tag{7}$$

Here, $v_{W,M} = 0.39$ mL(H_2O) / mL(tissue) is the volume fraction of water in myelin and $v_{W,NM} = 0.86$ mL(H_2O) / mL(tissue) is volume fraction of non-myelin water. While Jung et al. reported improved regression results when employing arguments based on geometric properties of myelin ($v_{W,M} = 0.36$ mL(H_2O) / mL(tissue)), we utilized the above mass-density based approach to avoid assumptions regarding the number of myelin lamellae in our more severely hypomyelinating mouse strains.

Axon volume fraction (AVF) maps were computed by assuming our diffusion acquisition minimized contributions of myelin to the diffusion-visible microstructure within a voxel, i.e. that $AVF = (1 - MVF) \times AWF$. Finally, the MRI g-ratio maps were computed according to the formalism developed by Stikov et al. (2015):

$$g_{MRI} = \sqrt{1 - MVF/(MVF + AVF)} \tag{8}$$

Following computation of quantitative parameter maps, the maps were registered to a high resolution ($40 \times 40 \times 40$ μm^3) T_2-weighted anatomical template of the mouse brain (Dorr et al., 2008) using a multi-step process. First, to overcome the deficit of limited WM contrast exhibited by the more hypomyelinating mouse strains, the echo time-axis averaged mGRE images were contrast-enhanced using Contrast Limited Adaptive Histogram Equalization (Stark, 2000). They were then bias field corrected using the N4 algorithm (Tustison et al., 2010). A series of rigid, similarity, affine, and symmetric diffeomorphic transformations were then obtained by mapping the contrast-enhanced and corrected mGRE images in native space to the template space using software implemented in Advanced Normalization Tools (ANTs) software (Avants et al., 2011). These transformations were then applied to all dwGRASE- and mGRE-derived parameter maps. This produced parameter maps for every mouse brain in template space and allowed comparison across individuals *and* mouse strains. Parcellation of white matter tract regions of interest (ROIs) was achieved using atlas labels derived from the Allen mouse brain atlas (Lein et al., 2007). The ROIs selected for study included the arbor vitae of the cerebellum (102,836 voxels; 6.58 μL), the internal capsule (28,667 voxels; 1.83 μL), the corpus callosum body (98,343 voxels; 6.29 μL), the corpus callosum genu (12,439 voxels; 0.80 μL), and the anterior commissure (13,020 voxels; 0.83 μL). These ROIs were selected because they were prominently defined and spanned most of the brain along the rostro-caudal axis.

3.4. Immunohistochemical staining

Five brains were used for immunohistochemical investigation: one each of the *KI, WT, 1xKO, 2xKO*, and *3xKO* genotypes. IHC was not collected for *Shi* strain as this genotype is known a-priori to be devoid of MBP expression. Brains were fixed overnight in 4 % paraformaldehyde at 4 °C. Brains then underwent paraffin embedding and were sectioned coronally at a thickness of 5 μm. Care was taken to select coronal sections that had the best possible level-matching along the rostro-caudal axis between brains. IHC was performed using the Ventana Discovery Ultra System (Roche, Laval, QC) with two antibodies, Myelin Basic Protein (Cell Signalling #78896S) and Neurofilament-L (LNF, Cell Signalling #2837S). Briefly, slides were deparaffinized (EZ prep buffer), followed by cell conditioning using the reagent CC1 at 95 °C, with variable incubation time for each antibody (LNF 32 min, MBP 24 min). The slides then underwent blocking with Inhibitor conditioning module (CM), for 8 min at 37 °C. Slides were incubated with primary antibody for one hour at 37 °C at the appropriate dilution (LNF & MBP 1:200). Next, 1 mL of secondary antibody, OmniMap anti-Rb HRP was applied to slides for 16 min. 1 mL of H_2O_2 CM was applied for 4 min, then one drop of DAB (diaminobenzidine) CM was added for 8 min to reveal staining. Finally, 1 mL of Copper CM was applied for 4 min. Counterstaining was performed by adding 1 mL of Hematoxylin to the slide for 16 min and subsequently adding 1 mL of Bluing Reagent for 4 min. Slides underwent dehydration in increasing concentrations of ethanol (95 %, 2 × 100 %) followed by xylene (2x). The resulting slides were mounted with acrytol mounting media and imaged at 20x magnification (0.25 μm/pixel) using an Aperio ScanScope CS optical microscope system (Leica Biosystems, Lincolnshire, IL).

3.5. IHC data analysis

Following high resolution microscopy, coronal plane IHC images were registered to a pre-selected coronal slice of the corresponding MRI template. Registration was carried out using an image processing pipeline developed in-house. Briefly, each IHC image was first downsampled to the in-plane resolution of the coronal MRI template slice. Subsequently, IHC images were decomposed from a red, green, and blue (RGB) color channel representation to a Hematoxylin (blue) and DAB (brown) component representation. This decomposition was carried out using the color deconvolution algorithm proposed by Ruifrok and Johnston (2001). This algorithm yielded a virtual DAB (vDAB) chromogen color channel, which quantified the DAB intensity and therefore the associated binding of the primary antibody to its target antigen. Next, the original RGB IHC image was binarized and inverted to highlight gray matter (GM) / white matter contrast. The resulting image was intensity thresholded to achieve a zero background. Following these preprocessing steps, rigid, affine, and symmetric diffeomorphic transformations were obtained using ANTs software to map the grayscale representation of the IHC image to the MRI template. These transformations were then applied to the deconvolved Hematoxylin and DAB components images. Finally, the Hematoxylin and DAB images were reconstituted into their original RGB representation in MRI template space to verify anatomical alignment and the vDAB image was used to evaluate MWF/MBP and AWF/LNF correspondences. To evaluate the correlation of MRI and histologically derived measures of WM microstructure, a scatter plot was generated within each ROI studied. An outlier rejection method, which excluded voxels deviating by >95 % from the centroid of points of the scatter plot for each strain was applied to account for errors due to misregistration of the IHC-derived images to those from MRI. Prior to regression analysis, all data points in scatter plots were normalized to the centroid of the *WT* cluster to facilitate quantitative comparison between WM tracts which originated from different coronal sections along the rostro-caudal axis.

4. Results

4.1. Odd/even echo misalignment correction for bipolar readout gradient acquisitions

Fig. 1 shows the evaluation of the proposed strategy on our mGRE data. Specifically, in Fig. 1**A**, the MI curves are plotted as a function of phase ramp slope, β, for five representative mouse brains. All curves exhibited a monotonic increase to the maximum value of MI and a subsequent monotonic decrease. This feature provides evidence that an optimal phase ramp slope, β^*, was calculated for each mGRE acquisition. To systematically quantify the impact of the proposed odd/even echo misalignment correction, we computed the lateral shift along the readout axis in the weighted centroid position of each mouse brain image, relative to the first echo. Fig. 1**B** visually demonstrates the impact of echo misalignment: the uncorrected multi-echo centroid displacements are shown in red, and the corrected values are shown in blue. Amelioration of inter-echo misalignment is evidenced by the absence of a jitter in the echo-time domain when the optimal phase ramp

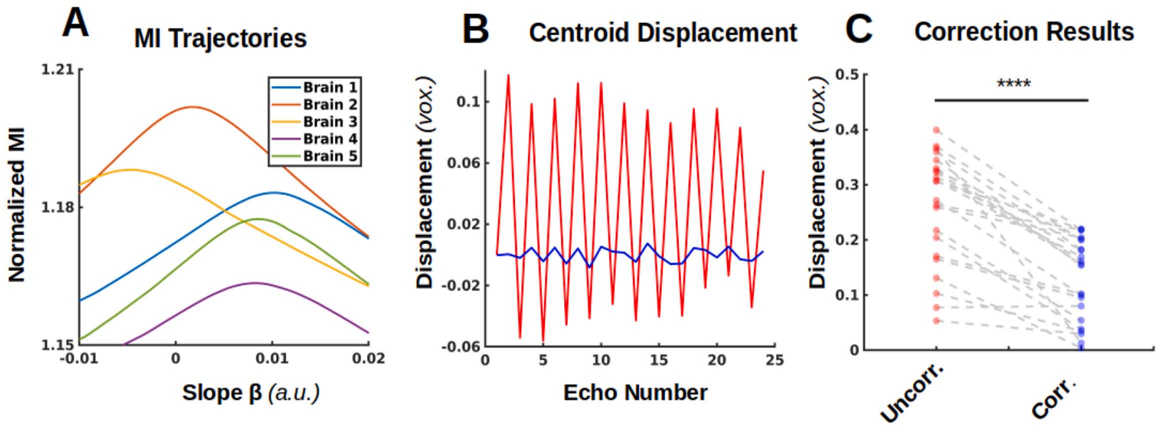

Fig. 1. Bipolar gradient correction strategy for precise MWF mapping. (A): Representative Mutual Information (MI) curves for 5 brains, plotted as a function of the applied phase ramp slope, β. (B): Representative jitter in the x-coordinate of the weighted centroid of the mouse brain along the readout direction (red trace) and its amelioration upon correction (blue trace). (C): Impact of odd/even echo misalignment correction on the average *x*-coordinate displacement for all brains included in this study. The four asterisks indicate a statistically significant difference ($p < $ 1e-4) in centroid displacement for the uncorrected (red) and corrected brains (blue).

slope β^* was applied. To evaluate the average effect of misalignment correction on our entire mouse cohort, we computed the mean inter-echo centroid displacement pre- and post-correction for each brain. Fig. 1**C** shows that, in all but one brain, the proposed correction decreased the odd/even echo misalignment in the whole cohort from ~0.3 voxels (~30 μm, red dots) on average to an average of ~0.15 voxels (~15 μm, blue dots). Overall, a statistically significant decrease ($p < $ 1e-4, Student's *t*-test) in misalignment was observed for all the mGRE images in our study.

4.2. Evaluation of LDR-rPCA for ex vivo myelin water fraction mapping at 7T

Following odd/even echo misalignment correction, we applied LDR-rPCA for ex vivo, mouse brain MWF mapping. We first examined the time courses of T_2* decay for the source signals, *M,* (raw mGRE data) and the two unit-rank components L_1 and L_2 obtained from LDR-rPCA for a representative wild-type mouse brain (Fig. 2). Fig. 2**A** demonstrates that the rapidly decaying L_2 component exhibited the expected, accelerated T_2* decay, representative of the myelin water pool. The L_2 component also showed a greater absence of signal at later echoes in the mGRE readout (Echo 16 and Echo 24 are shown). The sparse component, *S*, exhibited very little signal intensity, even in the first echo, indicating artefacts associated with odd/even echo misalignment were decreased through our chosen pre-processing. A granular examination of a selected voxel in the arbor vitae of the cerebellum (voxel location demarcated by grey, yellow, and blue arrows; Fig. 2**A**) demonstrated that the time course of each source signal behaved in the expected manner (Fig. 2**B**); the rapidly decaying component L_2 was almost completely absent by echo 15 (32 millisecond echo time). Meanwhile, L_1 and *M* exhibited a more slowly decaying profile and converged in magnitude at later echo times. The latter observation supports that the rapidly decaying component, corresponding to myelin water signal, has been identified from the corresponding source signal, *M*.

An example of voxel-wise mono-exponential fitting of the *M,* L_1 and L_2 components is given in Fig. 2**C**. The fits reveal relatively uniform proton density (*PD*) calculated from the source signal, *M*. By contrast, the *PD* map measured for the L_1 component exhibits hypointensity in white matter. The *PD* map calculated for L_2 component delineates the major mouse brain myelinated white matter tracts. Whole-brain histograms of the T_2* components measured from LDR-rPCA (Fig. 2**D**) demonstrated a clear separation of T_2* values of the two unit-rank components (mean T_2* in L_1 = 45.3 ms; mean T_2* in L_2 = 19.7 ms). This indicates our chosen implementation of linear dimensionality reduction for MWF mapping successfully extracted mGRE signal originating from myelin water protons (Fig. 2**D**).

4.3. Regional myelin water fraction values in MBP enhancer-edited mouse brains

MWF maps, as well as selected white matter regions of interest (ROIs) in the mouse brain are shown Fig. 3. Specifically, Fig. 3**A** shows our five selected white matter ROIs overlaid on a high-resolution template brain (40-micron isotropic resolution). The white matter ROIs include the arbor vitae of the cerebellum (*av*; red), the corpus callosum body (*cc(b)*; violet), the corpus callosum genu (*cc(g)*; gold), the anterior commissure (*ac*; plum), and the internal capsule (*ic*; cyan). Fig. 3**B** shows example structural mGRE images registered in the high-resolution template space. Specifically, genotype-averaged structural images for the *WT* ($n = 5$) and severely hypomyelinating mouse (*2xKO*; relative *Mbp* mRNA = 28 %; $n = 3$) are displayed. The *2xKO* structural image exhibited considerably diminished WM / GM contrast when compared to the *WT* image, as expected. Both the *WT* and *2xKO* average structural brain images also exhibited excellent spatial alignment with the anatomical template. Notably, the spatial alignment was improved by computing registration transformations using brains with artificially enhanced WM contrast (described in Materials and Methods). MWF hypointensity in the *2xKO* brain relative to *WT* was observed in all WM tracts (Fig. 3**C**).

Fig. 4 displays the MRI-derived MWF values in each white matter ROI and the corresponding correlation of MWF with *Mbp* mRNA. Comparison of the mean MWF of each mouse strain in each ROI indicated a statistically significant difference in MWF between the *WT* and all hypomyelinating mouse strains (relative *Mbp* mRNA < 100 %, Fig. 4**A**). The statistical comparison was conducted using Tukey's honest significance difference (HSD) test. Comparison of the mean MWF of the severely hypomyelinating (*2xKO* and *3xKO*) mouse strains to the *Shi* brains did not yield statistically significant differences. Moreover, no significant differences were observed between the *WT* and the *KI* strains in any ROI. Fig. 4**B** explores the relationship between MWF and *Mbp* mRNA within each ROI. Each subplot shows a clear linear relationship (blue line) ± 95 % confidence intervals (light blue shade). A statistically significant correlation (Pearson's $r > 0.85$, $p < $ 1e-4) was observed between MWF and *Mbp* mRNA in all white matter ROIs. Since there was no statistically significant difference between *WT* and *KI* conditions, the *KI* measurements were excluded from the regression analyses.

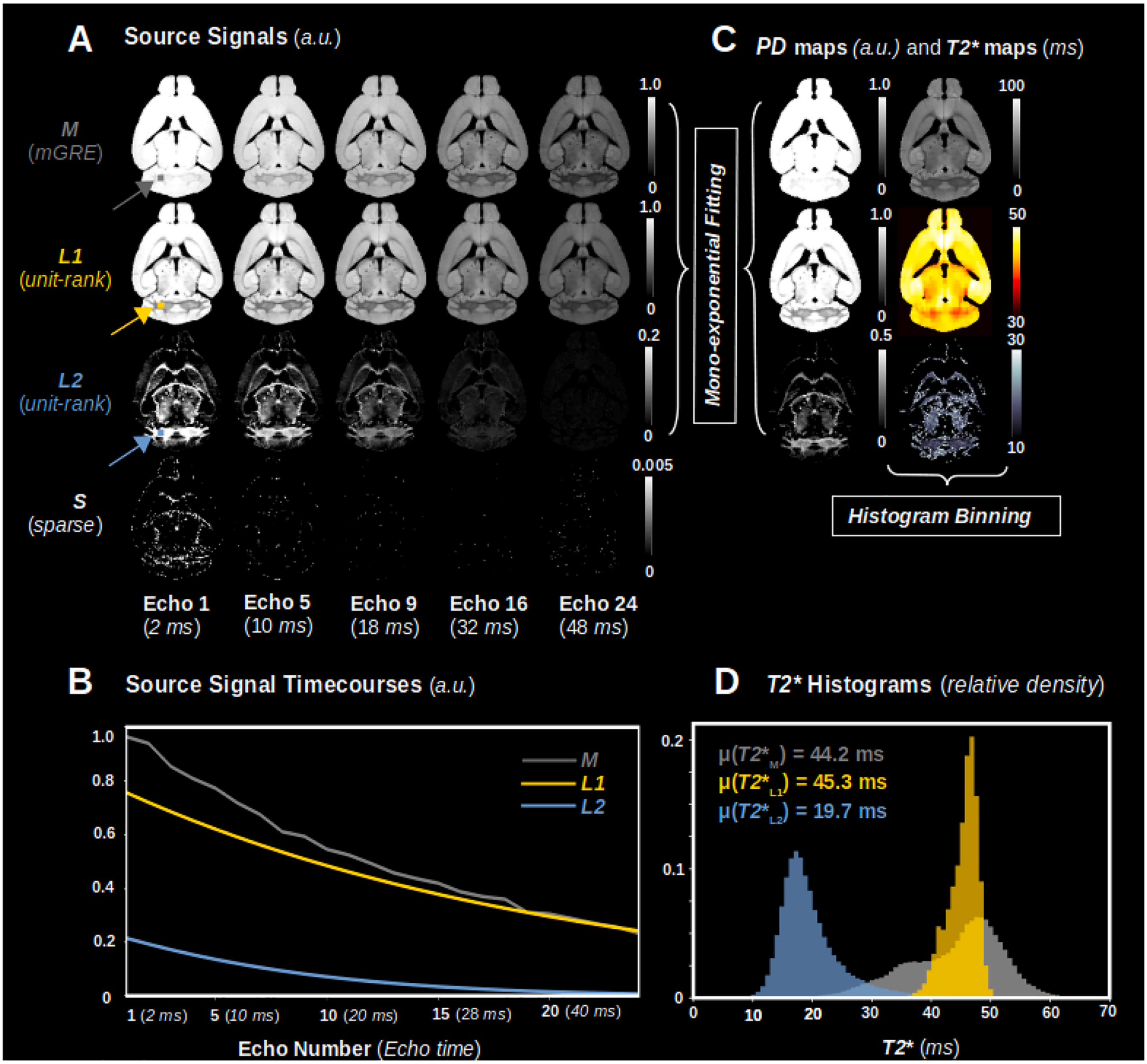

Fig. 2. Application of LDR-rPCA for MWF mapping. The example data in Fig. 2 is from a single, representative *WT* mouse brain. (A): Source signal time evolution as a function of echo number. (B): Source signal decay curves computed from LDR-rPCA for a selected imaging voxel in the cerebellar WM. The specific ROI is denoted by the gray, yellow, and blue arrows in Fig. 2A. The rapidly decaying L_2 component is characteristic of myelin water. (C): Maps computed from mono-exponential fitting of *M,* L_1 and L_2 components. The visible hypointensity in the L_1*–derived PD* map and the corresponding hyperintensity in L_2 *PD* map demarcate myelinated WM in the mouse brain (D): Histogram binning of T_2* maps for each component reveals a clear separation of T_2* values between L_1 and L_2. The average T_2* value for each distribution is provided within the panel.

4.4. Correspondence of myelin water imaging with myelin basic protein IHC

In order to establish a ground truth measure of MBP expression and myelin elaboration to compare with our findings from MRI, we performed immunohistochemical assays for MBP. To facilitate a quantitative comparison between MBP IHC microscopy slides and MRI-derived MWF, the intensity of diaminobenzidine (DAB) staining for MBP was deconvolved from the original red/green/blue representation of the IHC image and co-registered to a coronal slice of the anatomical template, yielding a "virtual" DAB (vDAB) image. The visual correspondence between MRI-derived MWF and MBP IHC maps is displayed in Fig. 5. Fig. 5**A** shows the approximate slice position of our chosen WM ROIs along the rostro-caudal axis of the mouse brain template. The WM ROIs used to parcellate the MWF and co-registered MBP IHC images are analogous to those in Fig. 3**A**. They utilize an identical color scheme and set of abbreviations. Fig. 5**B**, **C**, and **D** explore the relationship between MRI- and IHC-derived measures of myelination in rostral-most, caudal-most, and intermediate 2D slices, respectively. In each of Fig. 5**B**, **C**, and **D**, the two left-most columns display group averaged structural MRI (mGRE) and quantitative MWF maps. The two right-most columns show the corresponding co-registered high resolution MBP vDAB and MBP IHC intensity images. A progressively diminishing WM contrast in the mGRE images was observed concurrent with progressive reductions in relative *Mbp* mRNA. A global decrease in MWF was also observed in the hypomyelinating mouse strains with a corresponding decrease in MBP

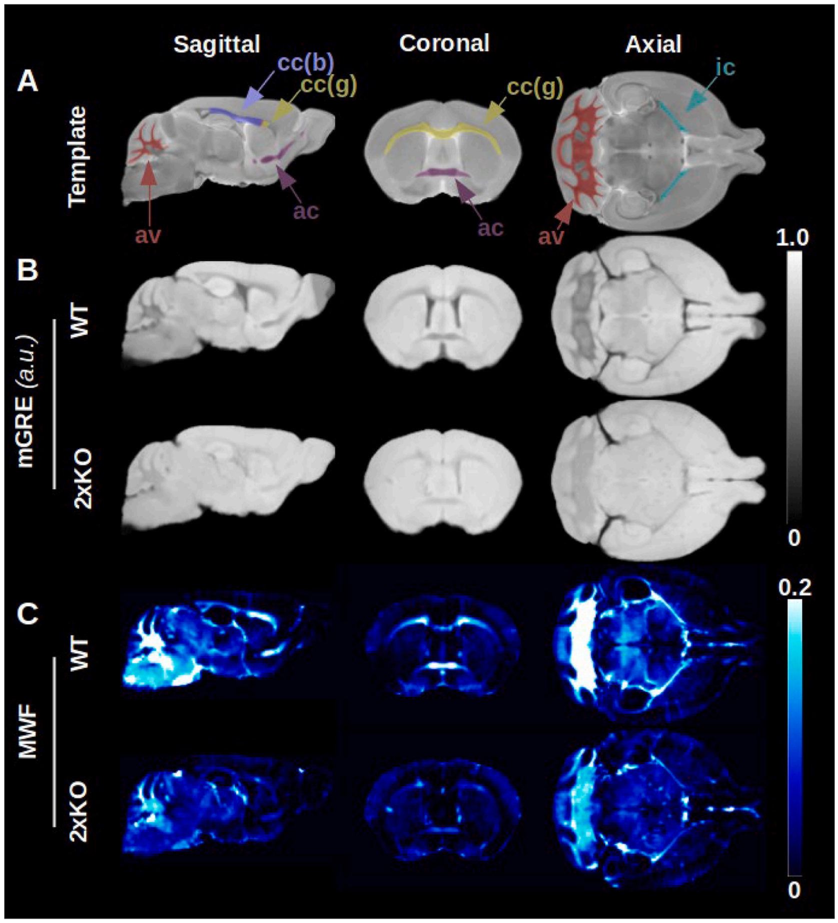

Fig. 3. Anatomical parcellation of the mouse brain and representative MWF maps. (A) 3D White matter ROIs investigated as part of our study overlaid on the anatomical template (*av*: arbor vitae; *cc(b)*: corpus callosum body; *cc(g)*: corpus callosum genu; *ac*: anterior commissure and *ic*: internal capsule). (B) Representative echo-time axis averaged structural mGRE images for the WT and 2xKO brains. (C) Corresponding MWF maps.

vDAB intensity.

The linear correlation between relative MWF and relative MBP vDAB intensity in WM ROIs is depicted in Fig. 5**E**. Each data point in the scatter plots of Fig. 5**E (i.-v.)** represents a single imaging voxel. A high, statistically significant correlation between MWF and MBP vDAB (Pearson's $r > 0.70$ and $p <$ 1e-4) was observed in all ROIs. In general, voxels in the scatter plots clustered according to mouse strain of origin. Strains with the least *Mbp* mRNA clustered in the lower left quadrant, while those with most *Mbp* mRNA clustered in the upper right quadrant. The marginal distribution plots accompanying each genotype identify the degree of voxel-wise overlap between MWF and MBP vDAB intensity. The order of appearance of each marginal distribution along its respective axis corresponds well with the order of increasing mouse strain *Mbp* mRNA, thereby recapitulating a strong correlation between MWF and MBP vDAB intensity. There was substantial overlap between the KI and WT conditions, as expected from the MWF parcellation results in Fig. 4**A**.

4.5. Diffusion imaging and correspondence with light chain neurofilament IHC

To evaluate the possibility that decreases in MWF signal may have been accompanied by decreases in axonal density, we carried out complementary diffusion-weighted imaging of each mouse brain included in our cohort. We specifically employed the NODDI model to estimate the axon water fraction (AWF) in all the WM tracts we studied with T_2*-based relaxometry. Fig. 6 shows MRI-derived AWF in white matter ROIs and the correlation of AWF with *Mbp* mRNA. The mean AWF in each ROI ± 95 % confidence interval is shown in Fig. 6**A**. Multiple comparisons of means from different mouse strains in each ROI indicated few statistically significant differences in axonal water fraction between the *WT* and mutant mouse strains. The statistical comparison was conducted using Tukey's HSD test, as was the case for the tract-wise comparison of MWF values shown in Fig. 4**A**. A statistically significant difference in axonal water fraction ($p < 0.05$) between *Mbp KO* mouse white matter and *WT* white matter was only detected in the anterior commissure for the *3xKO* strain (Fig. 6**A (i.)**) and corpus callosum body and genu for the *Shi* strain (Fig. 6**A (ii.)** and **(iii.)**). In all other cases, no significant differences between *Mbp KO* and WT AWF were observed in any ROI or mouse strain. Linear regressions of AWF vs. *Mbp* mRNA are shown in Fig. 6**B**. For all WM ROIs, a moderate linear correlation was observed between AWF and *Mbp* mRNA (Pearson's *r* in ~ [0.3, 0.5]). In particular, statistically significant ($p < 0.05$) correlations were observed in the corpus callosum genu, corpus callosum body, and internal capsule (Fig. 6**B (ii. - iv.)**). However, the estimated slope of the AWF vs. *Mbp* mRNA regression remained shallow and comparable to the estimate of its uncertainty at the 95 % confidence interval - indicating a very weak dependence, if any, of AWF on *Mbp* mRNA.

To supplement diffusion imaging-derived metrics of axonal density with histology-derived measures, we carried out IHC for light chain neurofilament (LNF). Neurofilaments (NFs), specifically the neurofilament light chains (LNFs), serve as crucial structural elements within the cytoskeleton, exclusively present in neurons and axons. These proteins, abundant in large myelinated axons of both the central (CNS) and peripheral (PNS) nervous systems, hold promise as biomarkers for assessing neuronal degeneration (Zanella et al., 2022). The visual correspondence between MRI-derived AWF and LNF IHC maps is displayed

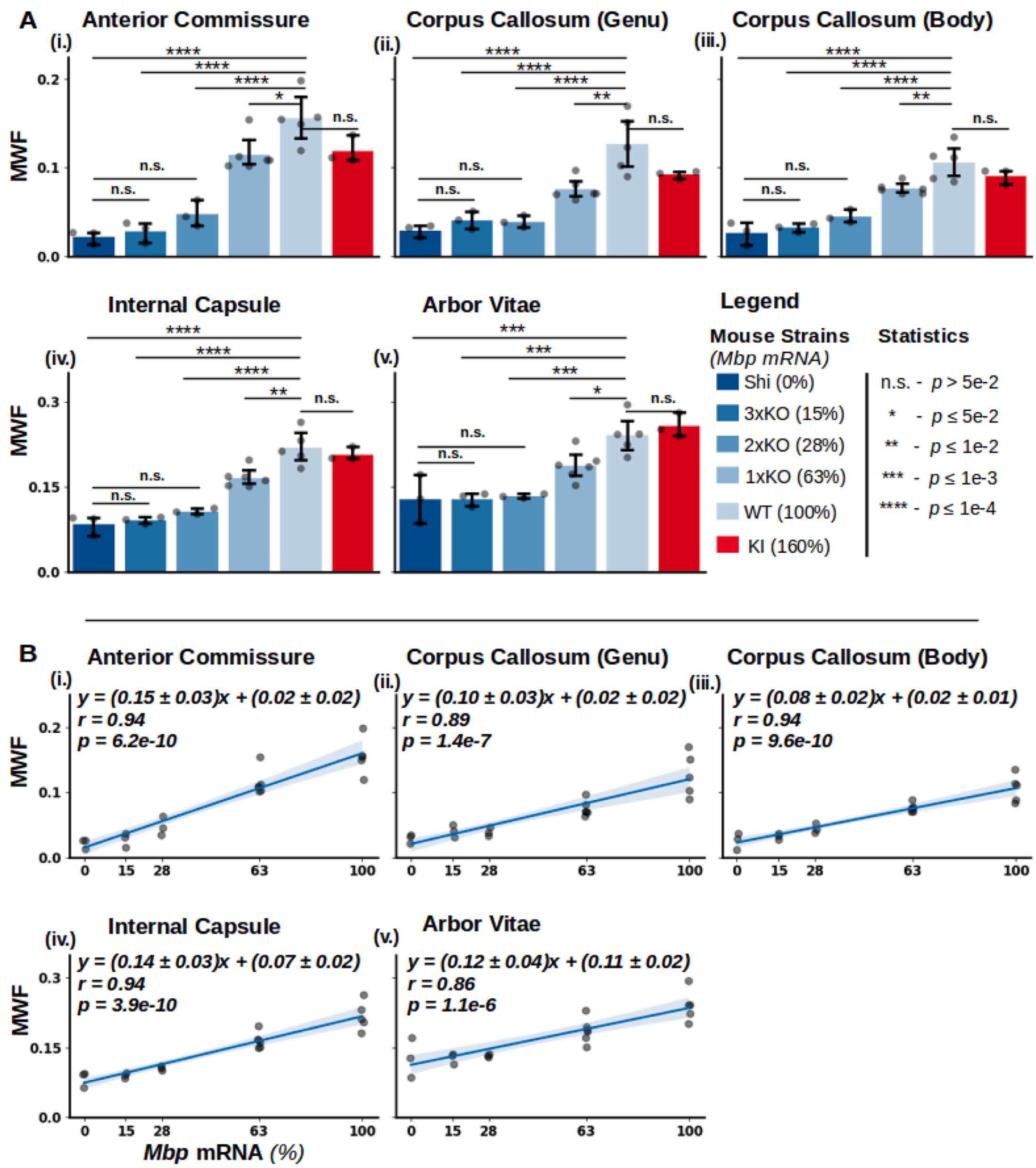

Fig. 4. MRI-derived MWF in selected white matter ROIs and correlation of MWF with Mbp mRNA. (A): Mean MWF for each mouse strain in selected white matter ROIs (error bars indicate ±95 % confidence intervals). Each data point indicates a measurement from an individual mouse brain. The legend indicates the mouse strains (relative *Mbp* mRNA % in brackets). The number of asterisks describes the degree of significance. (B): Relationship between *Mbp* mRNA and MWF (blue line) in each white matter ROI (95 % confidence intervals are shown in light blue). The equation for the line of best fit is shown in each subplot, with corresponding Pearson's correlation coefficient (r) and p-value. All linear fits were statistically significant ($p < 0.05$).

in Fig. 7. Fig. 7**A** shows the approximate slice position, along the rostro-caudal axis of the mouse brain, of our chosen WM ROIs. Fig. 7**B, C**, and **D** demonstrate the relationship between MRI- and IHC-derived measures of axonal density. These plots are analogous to those of Fig. 5**B, C**, and **D** for myelination. In each of Fig. 7**B, C**, and **D**, the two left-most columns display group averaged structural MRI (average of $b = 0$ diffusion-weighted images) and quantitative AWF maps. The two right-most columns show the corresponding co-registered high resolution LNF vDAB and LNF IHC intensity images. While there was a pronounced decrease in WM contrast as a function of decreasing *Mbp* mRNA in the averaged $b = 0$ images, the AWF maps did not show significant contrast variations between the graded hypomyelination phenotypes. LNF IHC exhibited a consistent staining pattern between genotypes, with LNF vDAB intensity exhibiting greater variation within subject than across genotype.

The linear correlation between AWF and LNF vDAB intensity in WM ROIs is depicted in Fig. 7**E**. Each data point in the scatter plots of Fig. 7**E (i.-v.)** represents a single imaging voxel, analogous to the scatter plots of Fig. 5**E**. While a statistically significant correlation was observed for all ROIs, with the exception of the arbor vitae (Fig. 7**E (v.)**), Pearson's r was low and had values ranging between $\pm \sim 0.1$. Unlike the results obtained when studying MWF vs MBP IHC correlations, there was no clear clustering of AWF and LNF IHC voxels according to genotype. This was evidenced by the marginal distributions of both LNF vDAB and AWF which showed considerable overlap between genotypes. Further, the distributions of LNF vDAB and AWF did not naturally order from lowest to highest *Mbp* mRNA levels. Taken together, the above findings support that reduced *Mbp* mRNA and hypomyelination had limited impact on AWF and LNF IHC in our enhancer-edited mouse lines.

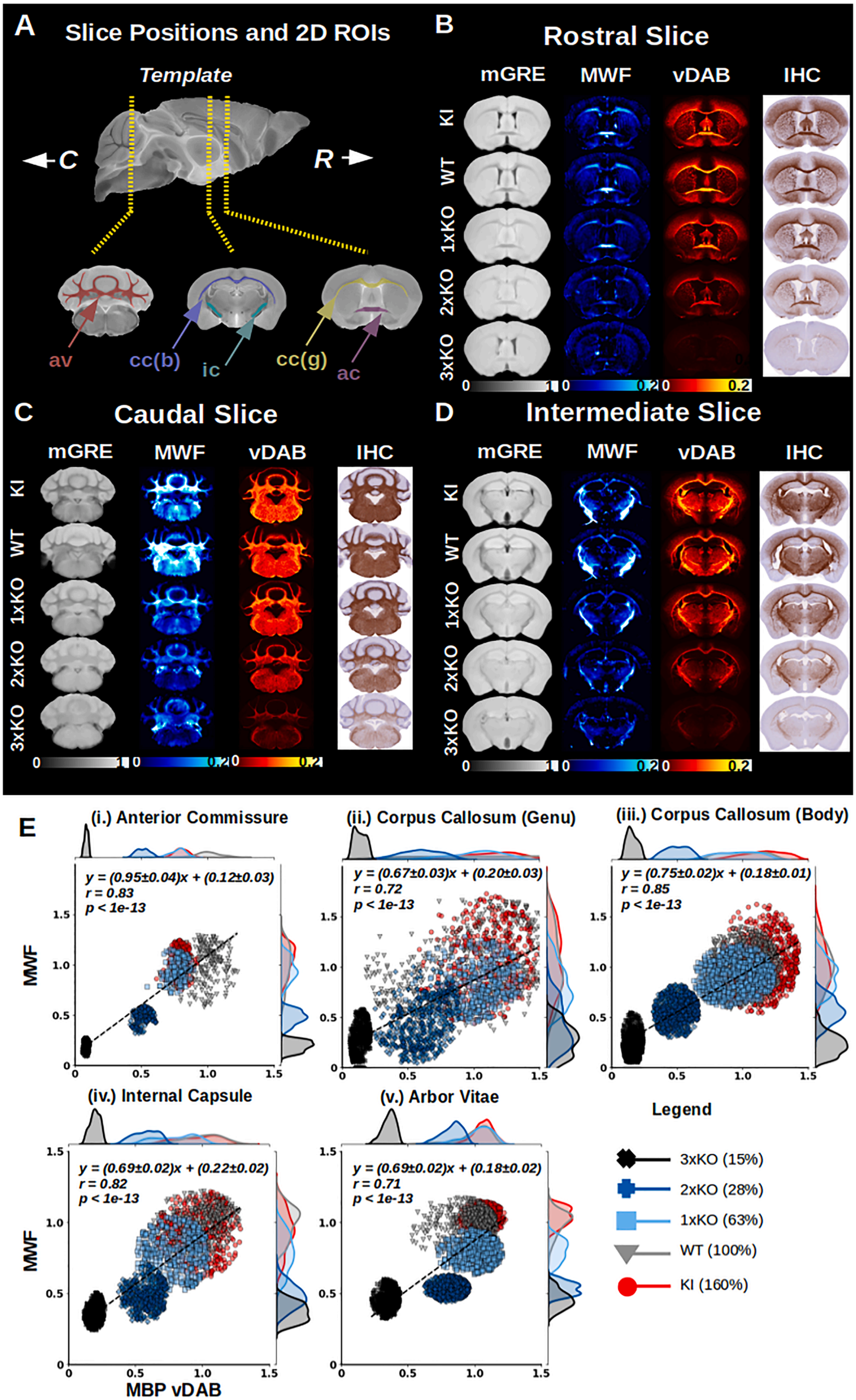

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Fig. 5. Correspondence between MRI-derived MWF and MBP IHC maps. (A): Approximate coronal slice location along the rostral-caudal axis with 2D ROIs derived from the template atlas. The 2D ROIs are identical to the 3D ROIs described in Fig. 3A but span only one slice. (B, C, D): MRI/Histology correspondence in rostral, caudal, and intermediate slices respectively. Group averaged structural MRI (mGRE) and quantitative MWF maps are displayed in the two left-most columns. They exhibit progressively diminishing WM contrast and decreasing MWF from the highest (*KI*, top) to lowest (*3xKO*, bottom) relative *Mbp* mRNA levels. This trend is corroborated by decreases in MBP vDAB intensity, as well as a diminishing DAB/Hematoxylin contrast in the two right-most columns of each panel. (E): Scatter plots showing the relationship between relative MWF and relative MBP vDAB intensity for all white matter ROIs. Each data point represents a voxel in the co-registered MWF and vDAB parameter maps. Note that the data points in each subplot were normalized to the average MWF and MBP vDAB intensity corresponding to the WT strain cluster. The line of best fit, Pearson's correlation *r* and *p*-value are also included in each subplot (with corresponding ±95 % confidence intervals). Marginal distributions accompanying each subplot depict the distributions of MWF and MBP vDAB intensities. The legend specifies the marker color denoting each mouse strain and the corresponding relative *Mbp* mRNA level.

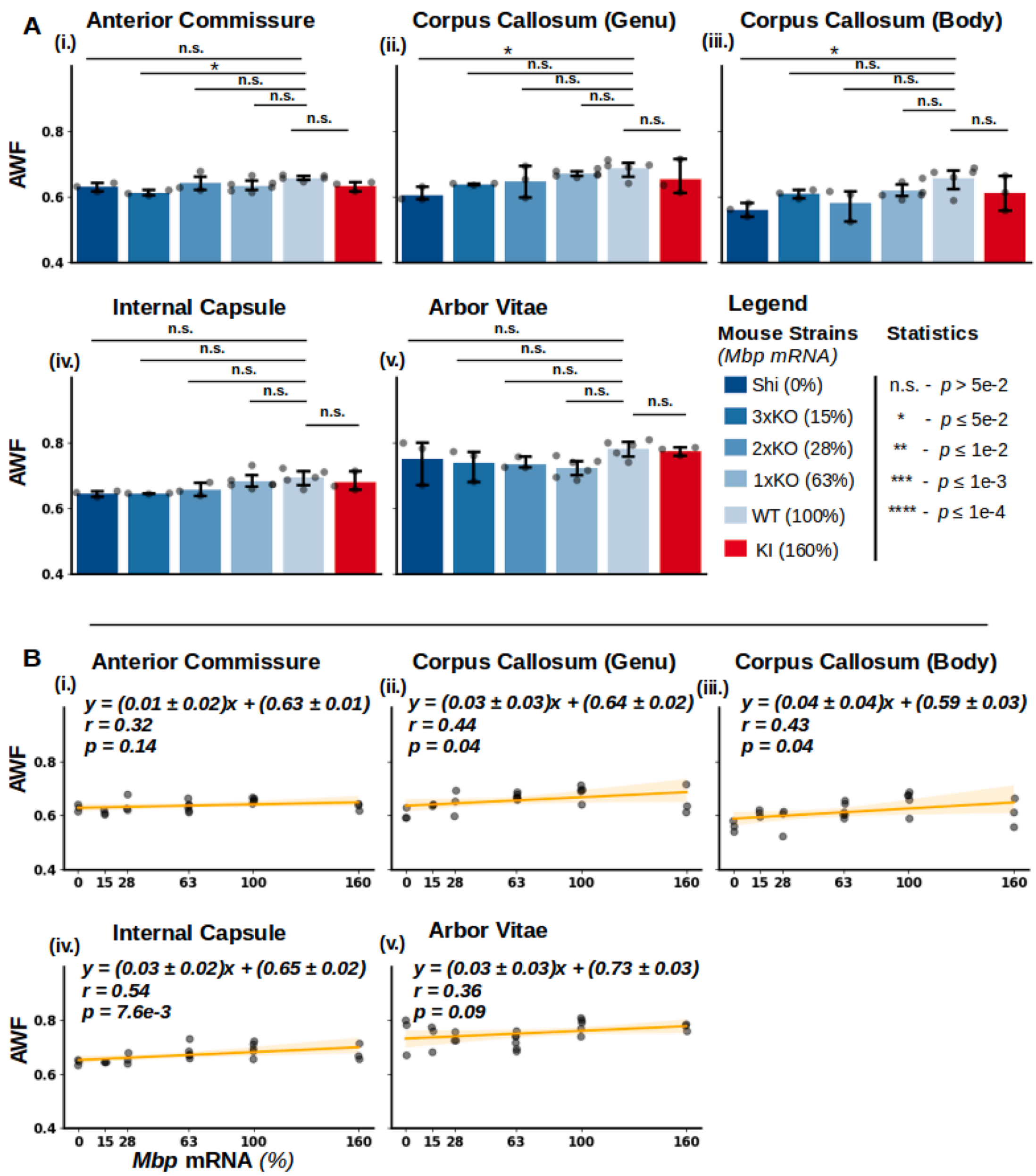

Fig. 6. MRI-derived AWF in selected white matter ROIs and correlation of AWF with *Mbp* mRNA. (A): Mean AWF for each mouse strain in selected white matter ROIs (error bars indicate ±95 % confidence intervals). The legend indicates the mouse strains (relative *Mbp* mRNA % in brackets). The number of asterisks describes the degree of significance. (B): Relationship between AWF and *Mbp* mRNA (yellow line) in each white matter ROI (95 % confidence intervals are shown in light yellow). The equation for the line of best fit is shown in each subplot, with corresponding Pearson's correlation coefficient (r) and p-value. All linear fits were statistically significant ($p < 0.05$) except for the anterior commissure ROI (subplot (i)). Linear fits were found to have very shallow slopes in all ROIs.

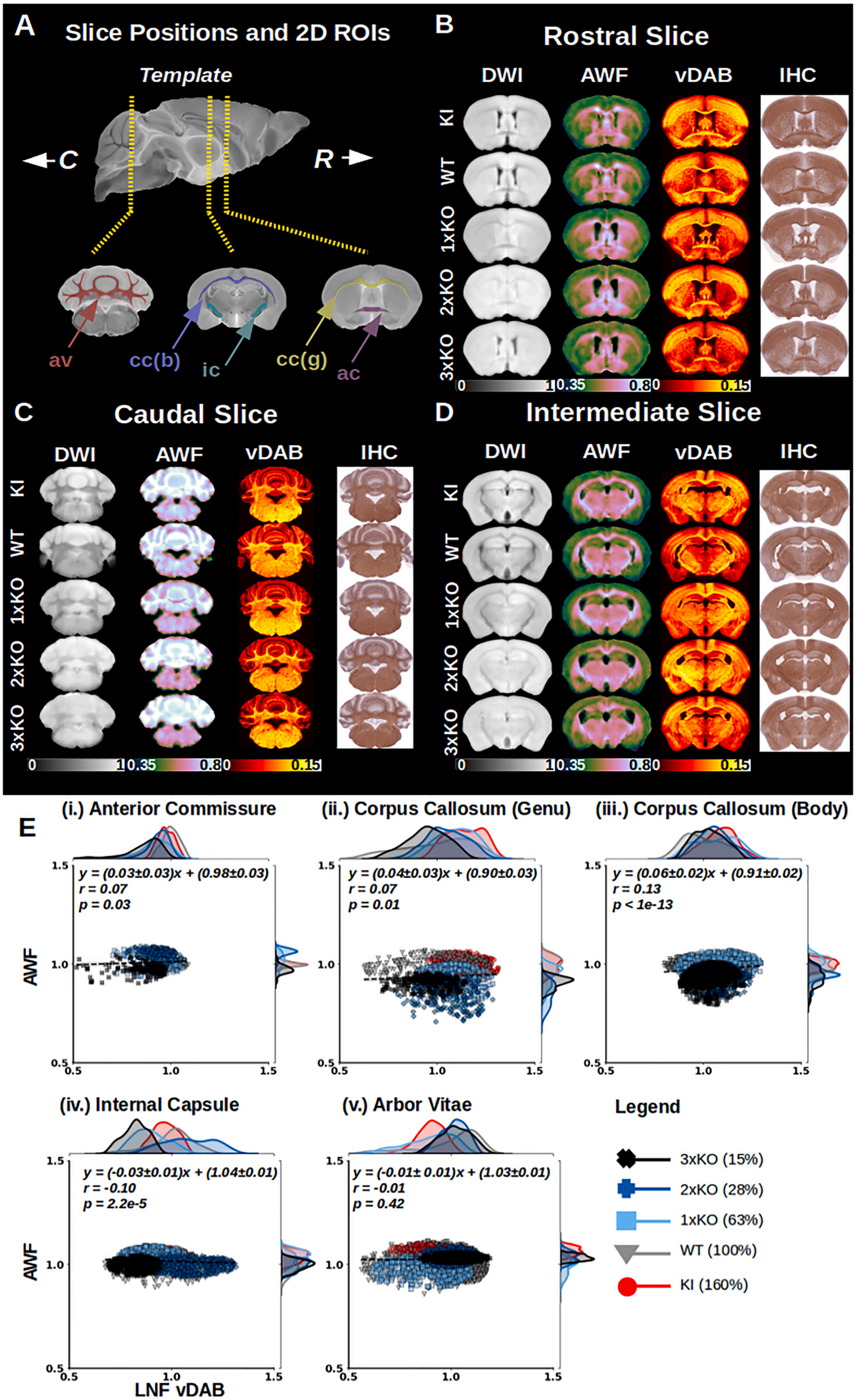

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Fig. 7. Correspondence between MRI-derived AWF and LNF IHC maps. (A): Approximate coronal slice location along the rostral-caudal axis with 2D ROIs derived from the template atlas. These 2D ROIs used are identical to the 3D ROIs described in Fig. 3A but span only one slice. (B, C, D): MRI/Histology correspondence in rostral, caudal, and intermediate slices respectively. Group averaged structural MRI (dwGRASE, $b = 0$) and quantitative AWF maps are displayed in the two left-most columns. The structural MRI images exhibit progressively diminishing WM contrast, yet consistent AWF when arranged from the highest (KI, top) to lowest (*3xKO*, bottom) relative *Mbp* mRNA levels. This trend is corroborated by the relative consistency in LNF vDAB and DAB/Hematoxylin intensities in the two right-most columns of each subplot. (E): Scatter plots showing the relationship between AWF and LNF vDAB intensity for all white matter ROIs. Note that the data points in each subplot were normalized to the average AWF and LNF vDAB intensity corresponding to the *WT* strain cluster. The line of best fit, Pearson's correlation *r* and *p*-value are also included in each subplot (with corresponding ±95 % confidence intervals). Marginal distributions accompanying each subplot depict the distributions of AWF and LNF vDAB intensities. The legend specifies the marker color denoting each mouse strain and the corresponding relative *Mbp* mRNA level.

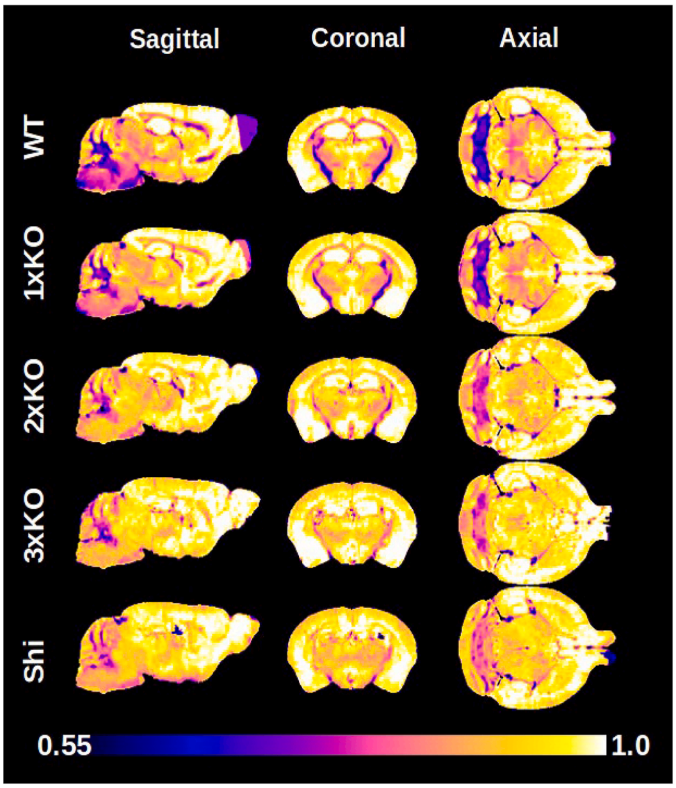

Fig. 8. Average MRI g-ratio maps, ordered from highest to lowest relative *Mbp* mRNA levels. Three slice planes are shown. MRI slice locations for the sagittal and axial planes are identical to those depicted in Fig. 3A. The coronal slice location is identical to that depicted in Fig. 5A. A diminishing contrast in aggregate g-ratio between white matter and gray matter, driven by increasing hypomyelination, is evident from top to bottom as g-ratio progressively increases in accordance with decreasing *Mbp* mRNA. Data from *KI* brains are not shown as they did not exhibit a statistically significant difference from the *WT* strain.

4.6. Magnetic resonance imaging of aggregate g-ratio

Utilizing information from the co-registered MWF and AWF maps, we computed whole brain aggregate g-ratio images for all the mouse brains in our study. Genotype-averaged, MRI-derived g-ratio maps are depicted in Fig. 8. A clear increase in white matter g-ratio, driven by increasing hypomyelination, is evident from top to bottom of Fig. 8. Moreover, the g-ratio image contrast between white matter and gray matter in the vicinity of the corpus callosum and anterior commissure was almost absent in the most hypomyelinating *Mbp KO* genotypes (bottom of Fig. 8). A precise quantification of MRI-derived g-ratio in selected white matter ROIs is provided in Fig. 9. Fig. 9**A** identifies that, for the WM tracts we studied, there was a statistically significant difference between *WT* g-ratio and the g-ratio of all hypomyelinating mouse strains. Analogous to the MWF measurements in Fig. 4**A**, we observed no difference in mean WM g-ratio between the *WT* and *KI* genotypes. We therefore excluded the KI line from Pearson's correlation analysis. We also identified a strong negative correlation between MRI g-ratio and *Mbp* mRNA (Pearson's r ∼−0.9, Fig. 9**B**). These results support our hypothesis that changes in MRI g-ratio are driven by graded decreases in *Mbp* mRNA and myelination in the *Mbp KO* mouse strains.

5. Discussion

The calculation of myelin volume fraction and g-ratio using non-invasive MRI measurements has shown significant promise for understanding neuroplasticity, neurodevelopment and neuroinflammation in the human brain. MWF and g-ratio mapping have the potential to non-invasively provide information about human brain microstructure that could otherwise only be derived from immunohistochemistry or broad field electron microscopy. However, to date, most MWF and g-ratio mapping studies have (i) applied a model that is not robustly validated relative to true myelin basic protein and myelination level, (ii) relied on lower spatial resolution measurements (thus not testing the limits of the technique) and (iii) restricted MWF and g-ratio mapping to a small number of white matter regions. In this work, we leveraged a panel of *Mbp* transcription enhancer-edited mice that exhibit graded hypomyelination profiles as an ideal testing ground for MWF calibration. Additionally, for the first time, we validated quantitative myelin water and g-ratio imaging using T_2*-based gradient echo MRI at the highest spatial resolution ever achieved.

Our acquisition scheme for high resolution MWF mapping with 7T MRI necessitated the implementation of high amplitude bipolar readout gradients. These are known to introduce sub-voxel shifts in magnitude mGRE data due to gradient timing delays and gradient waveform imperfections. Our proposed correction method, which accounts for contributions of the bipolar gradient-related artefacts in mGRE *k*-space, mitigated the impact of these sub-voxel shifts and ensured inter-echo alignment of the magnitude data (Fig. 1). Importantly, the bipolar correction method proceeded in a wholly data-driven manner. It can also be readily applied in conjunction with other existing correction strategies, such as slice profile compensation, that may be necessary with non-isotropic mGRE slice profiles in 2D imaging (Alonso-Ortiz et al., 2017).

Our chosen MWF computation method, LDR-rPCA, has advantages due to its model-free implementation and computational expediency. Instead of leveraging a biophysical model of WM, LDR-rPCA decomposes the total mGRE signal into two unit-rank components and a sparse component. The unit rank (L_1 and L_2) and sparse (S) components derived from LDR-rPCA represent slowly decaying, rapidly decaying, and artefactual components of the mGRE signal, respectively. Hankelization, which is part of the initialization in the LDR-rPCA formalism, ensures that the L_1 and L_2 components are unit-rank, therefore rejecting non-exponentially decaying components. Advantageously, non-exponentially decaying components are isolated in the sparse component map, S. This latter feature is attractive for application with mGRE data using long gradient echo trains, as later echoes may suffer from progressively decreasing SNR. Whole-brain MWF computations with LDR-rPCA typically proceeded in <8 min with our implementation, executed on a 16-core 2.10 GHz central processing unit (CPU) using Python linear algebra primitives. Further, since LDR-rPCA extracts myelin water and IEW pools by a series of matrix and vector operations on the 4D mGRE magnitude data, it is readily amenable to graphics

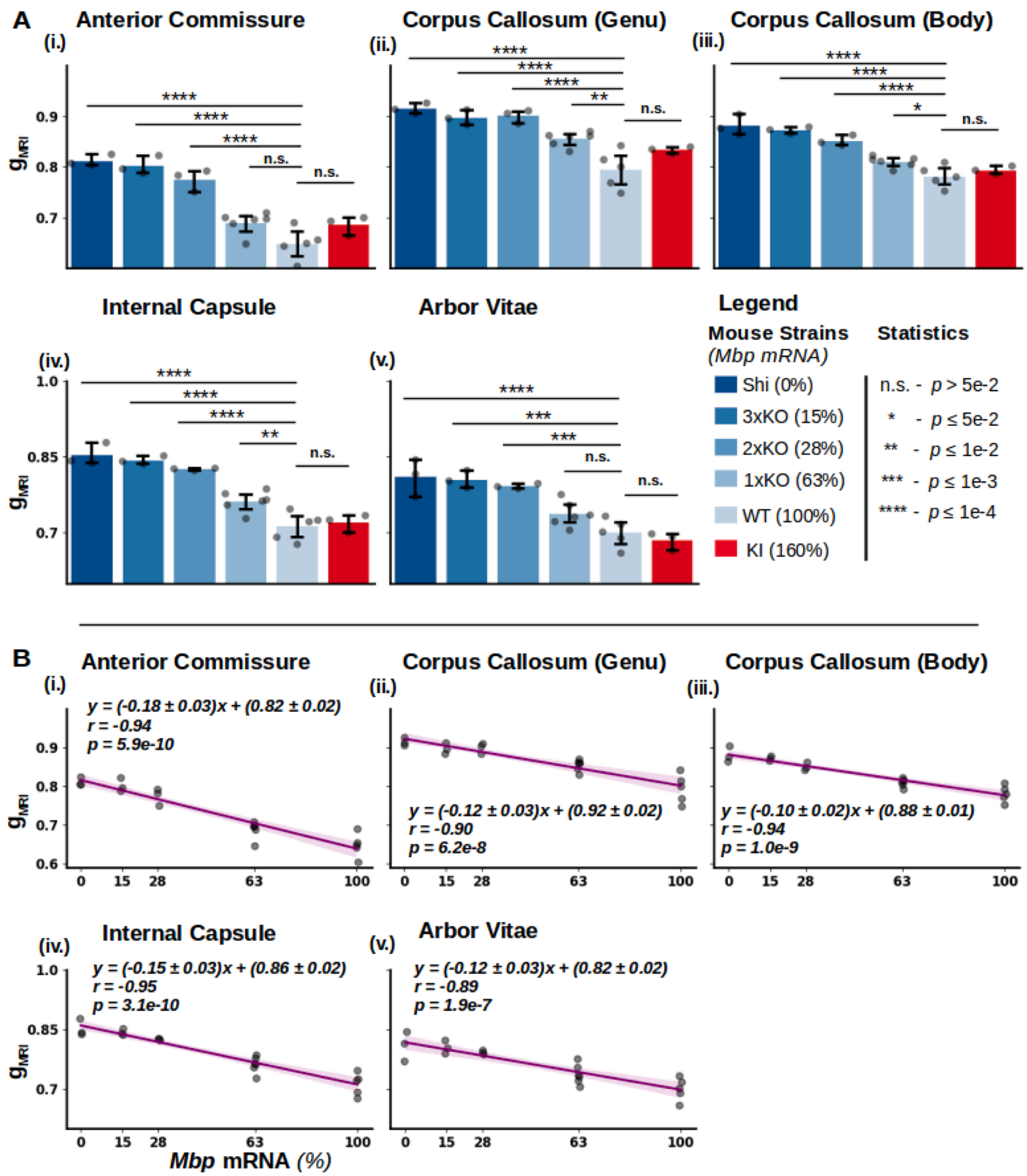

Fig. 9. MRI-derived aggregate g-ratio in selected white matter ROIs and correlation of g-ratio with *Mbp* mRNA. (A): Mean g-ratio for each mouse strain in selected white matter ROIs (error bars indicate ±95 % confidence intervals). The legend indicates the mouse strains (relative *Mbp* mRNA % in brackets). The number of asterisks describes the degree of significance. (B): Relationship between g-ratio and *Mbp* mRNA (purple line) in each white matter ROI (95 % confidence intervals are shown in light purple). The equation for the line of best fit is shown in each subplot, with corresponding Pearson's correlation coefficient (r) and p-value. All linear fits were statistically significant ($p < 0.05$) and exhibited an opposite trend to that observed for MWF (Fig. 4B).

processing unit (GPU) re-implementation. While not pursued in this study, GPU parallelization and consequent increased acceleration constitute an attractive avenue for future development.

Several other features supported our specific application of model-free LDR-rPCA for MWF mapping in the hypomyelinating mouse lines. During previous development of the method, Song et al. (2020) used in silico, virtual MWF phantoms to compare NLLS-derived three-pool MWF (Du et al., 2007) to LDR-rPCA MWF. NLLS has been applied by various groups but suffers from SNR constraints, as well as the inability of the often-employed trust-region reflective fitting algorithm to resolve redundant minima in the solution manifold space. The requirement to seed the NLLS algorithm with initial conditions and solution boundaries, which depend on tissue sample fixation status (in the case of ex vivo MRI) and magnetic field strength, can also limit the application of NLLS for "out of the box" MWF mapping in clinical studies (Liu, Grouza, et al., 2022). The in silico results of Song et al. previously demonstrated that LDR-rPCA was superior to NLLS in extracting true MWF under varying SNR regimes. Corresponding in vivo imaging experiments in 7 healthy human subjects also indicated high scan-rescan reproducibility and rejection of artefacts when applying the LDR-rPCA for MWF mapping. Finally, sensitivity to local demyelination was established by Song et al. using this method in a single patient with x-linked adrenoleukodystrophy, as evidenced by clinical diagnoses and the correspondence of MWF results with those obtained from quantitative inhomogeneous magnetization transfer (qihMT) imaging. Taken together, these findings offer encouraging support for the adoption of mGRE-based MWF mapping with LDR-rPCA.

For all the mouse brain samples included in our work, T_2* decay information was collected from glutaraldehyde fixed, ex vivo mouse brain tissue imaged at room temperature (21 °C). Fixed tissue is advantageous for high resolution MRI because it allows extensive signal averaging and improved SNR for myelin mapping. In this regard, our chosen spatial resolution afforded the ability to resolve small white matter fiber regions in the mouse brain. The mean T_2* values we measured in white matter tracts across the whole mouse brain for the myelin water and IEW pools were ∼20 and ∼45 milliseconds, respectively (Fig. 2**, Table S1**). These values differ markedly from those typically cited for in vivo conditions: ∼10 ms for myelin water and ∼60 ms for IEW at 7T (Shim et al., 2022). In general, T_2 and T_2* are known to shorten with increased magnetic field strength (Cohen-Adad, 2014) and aldehyde fixation (Dusek et al., 2019; Shepherd et al., 2009). Although less prominent at ultra-high fields, increasing the sample temperature is also known to moderately decrease transverse relaxation time (Birkl et al., 2016; Thelwall et al., 2006).

A limitation of our study, with relevance to comparisons to in vivo mouse tissue, is that it was conducted using fixed brains. In this regard, we hypothesize that the principal contributor to observed differences between our ex vivo measurements of the T_2* of the myelin and IE water pools and those obtained in vivo is the effect of aldehyde fixation. Specifically, aldehyde fixation is known to increase delamination and introduce intralamellar vacuoles into the myelin sheath (Seifert et al., 2019). This effect is likely exacerbated by destabilization of the lamellar structure of the myelin sheath because of depleted MBP. For fixed tissue, these factors likely contributed to a myelin water pool that was: (i) larger than in the in vivo case, thus increasing the measured MWF and (ii) characterized by a longer T_2* than in the case of in vivo imaging, owing to the less restricted molecular environment of the larger intralamellar spaces. In contrast, the effect of cross-linking proteins in the intracellular space may produce a more restricted molecular environment characterized by a shorter T_2* than that of in vivo tissue (Tayri-Wilk et al., 2020).

While we acknowledge the above limitations of the present results, it is also noteworthy that our MRI acquisition and model-free MWF computation method was consistently able to resolve the myelin and IEW components, even in the challenging conditions faced by our proposed solution to the inverse problem (Liu et al., 2022) (Fig. 2**D, Fig. S3,** blue and yellow histogram bars). Although the previously described effects of chemical fixation on white matter microstructure hinder the direct generalization of our findings to corresponding in vivo measurements, we believe the absence of physiological noise in our preparation also allowed us to maximally isolate the contributions of myelin and IE water pools to the measured mGRE signal.

A particular novelty of this study was the validation of MWF mapping sensitivity to myelin content by exploiting the differential myelination phenotypes of the *Mbp KO* mouse strains. This necessitated a customized, in-house implementation of the low rank MWF computation for application with our ultra-high-resolution ex vivo MRI dataset. We observed that LDR-rPCA produced high quality, biologically plausible MWF maps (Fig. 3). Increased WM MWF was observed in tracts that corresponded to anatomical WM priors from the Mouse Brain Atlas of the Allen Brain Institute (ABI) (Lein et al., 2007). Moreover, a pronounced global decrease in MWF was observed in all hypomyelinating mouse brains compared to *WT* mice, suggesting an overall sensitivity to decreased myelination (Fig. 4).

Based on our chosen MRI acquisition and processing pipeline, we sought to understand the regional impact of *Mbp* mRNA accumulation and myelin water fraction in specific white matter tracts. In all WM tracts studied, a graded decrease in MWF was observed as a function of *Mbp KO* mouse strain, demonstrating the previously established importance of *Mbp* expression on myelin elaboration and stabilization (Bagheri et al., 2024; Shine et al., 1992) (Fig. 4**A**). While we had the opportunity to parcellate an arbitrary number of WM tracts, we restricted our focus on those that (i) were well-delineated in the ABI atlas and (ii) had in-house corroborating MBP and LNF staining to provide ground truth IHC measurements for comparison. We believe this resulted in extraction of the most meaningful and generalizable measures of myelin volume fraction, axon volume fraction and MRI-derived g-ratio from our mouse panel. A robust correlation between *Mbp* mRNA and MWF was observed in all WM tracts studied (Fig. 4**B**). However, not all WM tracts we studied were equally affected by graded decreases in *Mbp* expression. Since cerebral myelination generally begins postnatally in the brainstem and proceeds rostrally, discrepancies in the tract-specific myelin and the corresponding regression models of MWF vs. *Mbp* mRNA may be attributable to preprogrammed spatiotemporal gradients of myelin elaboration in the developing mouse brain (Foran and Peterson, 1992). Indeed, we observed that the most caudal WM tracts (internal capsule and arbor vitae) exhibited a non-zero intercept in their regression plots vs. relative *Mbp* mRNA. This may suggest the existence of a compensatory myelination mechanism, active prior to P30, which is sensitive to disruptions in *Mbp* action. Advantageously, the regression models we obtained for each WM tract in the mouse brain can be utilized to calibrate the MWF signal in other pre-clinical murine studies of hypomyelination.

The strong correlations we observed between MWF and *Mbp* mRNA accumulation levels, as well as the close visual correspondence between MWF and digitized MBP IHC maps, solidify the understanding that *Mbp* expression, myelin elaboration, and MWF signal are closely related (Fig. 5). By carefully matching the selected coronal brain slices originating from different mouse strains based on their location along the rostrocaudal axis and registering them to the parcellated mouse brain template, we maximally isolated the impact of MBP-driven myelin elaboration on the MWF signal. This procedure also resulted in elimination of possible uncertainties typically associated with manual ROI delineation in MR images. Further, a comparison of our computed MWF values to those obtained in a previous high-resolution study using *WT* mice shows a close correspondence in several white matter tracts; the genu of the corpus callosum (West et al. MWF∼0.10±0.01 vs. our MWF=0.13±0.03), the body of corpus callosum (West et al. MWF∼0.11 ±0.02 vs. our MWF=0.11±0.02), and the temporal limb of anterior commissure (West et al. MWF∼0.11±0.02 vs. our MWF=0.09±0.02) (West et al. 2018a). Note however, the values in the West et al. (2018a) study were obtained from manual delineation of small ROIs comprising ∼12 voxels with 75 μm isotropic MRI spatial resolution. Ours were parcellated utilizing ROI masks derived from the ABI mouse brain atlas spanning thousands of voxels. Consequently, we believe our derived MWF values are, advantageously, representative of whole tract myelin measures. Our measurements from the KI mouse strain, with 160 % relative *Mbp* mRNA, did not yield an elevated MWF signal. Furthermore, the MWF values we measured from the KI mouse brain were consistent with a corresponding normal level of MBP IHC staining. This may be a result of post-transcriptional modifications which minimized *Mbp*-induced hypermyelination in the KI mouse at P30.

In order to evaluate whether the graded decreases we observed in MWF were truly attributable to graded hypomyelination and not decreased axon volume fraction in the *Mbp* KO mice, we performed complementary evaluations of bulk axonal density in all mouse brain WM tracts under investigation. Our chosen MRI measure of axonal density, termed the axon water fraction (AWF), has been used in previous studies (Wang et al., 2019). Importantly, we implemented a diffusion-weighted MRI acquisition scheme for AWF mapping that had a relatively long T_E (35 milliseconds) to reduce leakage of myelin signal into the biophysical modelling of axonal density. The bar plot representation in Fig. 6 demonstrates there are minimal *Mbp*-driven changes in AWF. A limitation of our chosen measure of axonal density is that AWF measures derived from the NODDI model are known to be partially driven by diffusion signal from larger axons. This could have affected our tract-wise AWF values. While NODDI has been criticized for overestimating the cerebrospinal fluid (CSF, v_{iso}) and intracellular (v_{ic}) volume fractions (Alsameen et al., 2023) due to the TE-dependency and differential T_2 weighting of these compartments (Gong et al., 2020), our ex vivo preparation exhibited minimal CSF contribution. Moreover, we expect that the T_2 of the intracellular and extracellular compartments are closer in value than would be expected in vivo secondary to the fixation process and the low temperature, as was observed for T_2* values obtained from the mGRE acquisition. However, we acknowledge that the use of additional calibration factors to scale NODDI-derived estimates of AWF may reduce the deviation from the ground truth axonal density (Papazoglou et al., 2024). The corroborating absence in variation of LNF staining between mouse strains studied supports the hypothesis that decreases in measured MWF were primarily a function of hypomyelination - not decreased axon volume fraction (Fig. 7).

Finally, we leveraged a biophysical model of WM water pools in conjunction with existing electron microscopy-derived data (Jung et al., 2018; West et al., 2018a) to compute MVF, AVF, and g-ratio measures in the whole mouse brain (Fig. 8). Given that (i) the effects of exchange on diffusion-derived measures of AVF are partially mitigated in aldehyde fixed specimens and (ii) our diffusion acquisition minimized contributions of myelin signal to measures of axonal density, we have confidence that our computed g-ratio values were closely related to the underlying white matter microstructure. Indeed, our values of aggregate g-ratio showed a tight correlation with *Mbp* mRNA (Fig. 9).

In summary, our work has demonstrated a robustly validated measure of MWF which leveraged a panel of CRISPR/Cas9-generated mouse lines displaying graded hypomyelination resulting from edits in the enhancers driving *Mbp* transcription. The joint application of LDR-rPCA with graded hypomyelination and availability of the *Mbp* enhancer-edited mouse lines permitted calibration of non-invasive MWF measures against direct measures of myelin elaboration. Our mGRE-based MWF mapping method was also integrated for the first time with the MRI g-ratio formalism, paving the way for calibrated whole brain g-ratio measurements in humans. We believe our method provides a robust MRI biomarker for the study of normal neurodevelopment, as well as pathology in the context of neuroinflammatory and neurodegenerative diseases.

Data and materials availability

All data needed to evaluate the conclusions in the manuscript are present in the manuscript. Sample data, analysis scripts, and the Python implementation of LDR-rPCA will be available at https://github.com/RudkoLab/mwf_rpca.

CRediT authorship contribution statement

Vladimir Grouza: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hooman Bagheri:** Writing – review & editing, Validation, Methodology, Investigation, Data curation. **Hanwen Liu:** Writing – review & editing, Software, Methodology, Investigation, Data curation. **Marius Tuznik:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Zhe Wu:** Writing – review & editing, Software, Methodology, Investigation. **Nicole Robinson:** Resources, Methodology, Investigation. **Katherine A. Siminovitch:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition. **Alan C. Peterson:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **David A. Rudko:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing interests relevant to this article.

Acknowledgments

V.G. and D.A.R. acknowledge I.R. Leppert, J.S.W. Campbell, and V.S. Fonov (McConnell Brain Imaging Centre, Montreal, Quebec, Canada) for helpful discussions regarding relaxometry, g-ratio imaging, and preclinical image registration.

Funding

V.G. is supported by the Fonds de Recherche du Québec Doctoral Training Award and the Multiple Sclerosis Canada endMS Doctoral Studentship. H.L. is supported by the Michael Smith Foundation for Health Postdoctoral Research Scholar Award and the Multiple Sclerosis Canada endMS Postdoctoral Studentship. H.B., K.A.C., and A.C.P. are supported by a Canadian Institutes of Health Research Foundation Grant (FDN 148457). Funding support was also provided by the National Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant (RGPIN-2018-05047), Multiple Sclerosis Canada Biomedical Research Operating Grant (Fund #3613) and the Canada First Research Excellence Fund (CFREF) Healthy Brains for Healthy Lives Innovative Ideas Project Grant of Dr. David A. Rudko.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2024.120850.

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